# Book strip

**Science · LAUNCH · Autumn 2 · Week 4 · SCI-LAUNCH-A2-W04-L2**

Use: from chromosome to gene to base

I can zoom from a chromosome to a gene and a base, and explain how the three levels fit together.

Success: label the three levels and explain a part–whole link.

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# Route card

**Staff: cut these apart before the lesson.** Give one route card. Other routes remain available.

## Supported

Supported

Supported route card with the tier name

Label the three zoom levels: chromosome, gene and base.

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

## Independent

Independent

Independent route card with the tier name

Explain what the gene bracket marks in the DNA drawing.

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

## Aspire

Aspire

Aspire route card with the tier name

Explain what is lost if a model labels an entire chromosome as a gene.

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

## Choice

Choice

Choice route card with the tier name

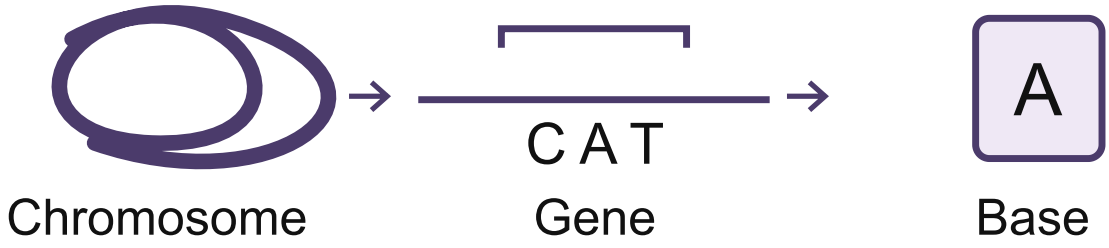
Which is packaged DNA: chromosome or base? Give a reason.

Use Reference A and B and the matching task sheet.

Staff I ☐ S ☐ C ☐

Made by Matt

# Reference A zoom drawing



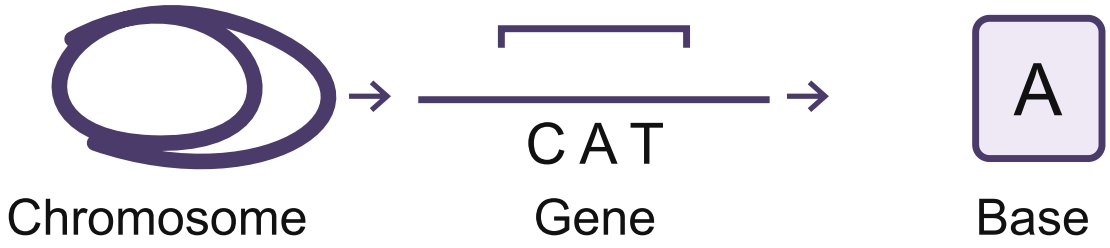

A coiled chromosome, a bracketed gene and the base A.

Reference only. The drawings are simplified and do not show scale.

| Level | What the picture shows |
|---|---|
| Chromosome | A long DNA molecule packaged with proteins. |
| Gene | A bracketed section of the DNA. |
| Base | One unit in the DNA sequence, shown here as A. |

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# Reference B explain the links




The three levels run from chromosome to gene to base.

Reference only.

| Link | Explanation |
|---|---|
| Chromosome to gene | A chromosome carries a long DNA molecule. One region of that DNA can be a gene. |
| Gene to base | The gene has a sequence of bases. A represents one base. |
| Model limit | The drawing leaves out packaging proteins and most bases. The sizes are simplified. |

The genome is the complete DNA set. It includes DNA from all chromosomes.

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# Catch-up sheet absence or access only

Use this after absence or when an access barrier prevents joining. It replaces a starting step.

1. A chromosome is a long DNA molecule packaged with proteins.

2. A gene is a section of DNA.

3. A, C, G and T are letters that represent bases.

Point to one level on Reference A and join the current model or first You do task.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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# Choice task sheet

Select an answer and give a short reason. Use Reference A and B.

**1. Which is packaged DNA: chromosome or base? Give a reason.** Staff I ☐ S ☐ C ☐

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**2. Which marks a section of DNA: gene or genome? Give a reason.** Staff I ☐ S ☐ C ☐

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**3. Which shows one base: A or the whole gene? Give a reason.** Staff I ☐ S ☐ C ☐

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